Certified recovery of trace-determinant structure under low-dimensional corruption

Daniel Eric Fredriksen
Quantyra Inc

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Abstract

Let an explicitly represented alternating trilinear tensor on $\mathbb{F}_2^{3s}$ be the sum of a nonzero trace-determinant tensor over an unknown field $\mathbb{F}_{2^s}$ and an arbitrary alternating perturbation supported on at most R binary covectors. We give a bounded randomized reconstruction using four contractions, scalar-field recognition, and a two-translate defect operator. When $s \geq 2R+5$, at most $8 \cdot 2^{4R} h$ trials return a verified compatible representation with probability at least $1-2^{-h}$. Every accepted result is checked exactly, including outside the sufficient promise. Conservative bit bounds are $\widetilde{O}(2^{4R} h n^{12})$ time and $\widetilde{O}(n^8 + \log(h+1))$ working space, where $n = 3s$. The result concerns supported algebraic corruption, not arbitrary noise or unique identification of a planted field. The companion contains executable finite checks. Complete Lean certification is in progress; this draft does not claim that the main theorem or the Python implementation is formally verified.

1 Problem and contribution

Put $V = \mathbb{F}_2^n$, $n = 3s$. An alternating trilinear tensor is given by the $d = \binom{n}{3}$ coefficients $C(e_i, e_j, e_k)$ for $i < j < k$, in lexicographic order. Trilinearity and alternation determine all other coefficients. Over $\mathbb{F}_2$, alternation implies symmetry, and coefficients with a repeated index are zero. This is a dense exact input model, of size $\Theta(n^3)$ bits; the input is not an oracle or a succinct circuit.

For a binary subspace $U \subseteq V^*$, write $U^\perp$ for its annihilator. We say E is supported on U when it factors through $V/U^\perp$ in every argument. Thus $E(x,y,z) = e(u(x), u(y), u(z))$ for a surjective binary linear map $u : V \to \mathbb{F}_2^r$ whose coordinate covectors form a basis of U, and an alternating trilinear form e. No distribution is assumed for U or e. This condition is different from a bound on tensor rank, a sparse set of corrupted entries, or independent statistical errors.

The proposed contribution is the specific worst-case supported-corruption guarantee below, with a complete certificate and charged search. The trace-determinant construction, adjoint equations, and cancellation of internal perturbations have substantial precedents; Section 7 describes the comparison. Neither priority nor a new general denoising principle is claimed.

Theorem 1 (Bounded compatible recovery)**.** *Let $0 \leq R \leq n$, $h \geq 1$ be integers. Suppose there exist a field $K \cong \mathbb{F}_{2^s}$, a binary linear isomorphism $f : V \to K^3$, $c \in K^\times$, and an alternating tensor E supported on U_* with $\dim_{\mathbb{F}_2} U_* \leq R$ such that*

$$C(x,y,z) = \mathrm{Tr}_{K/\mathbb{F}_2}\big(c \det(fx, fy, fz)\big) + E(x,y,z). \tag{1}$$

If $s \geq 2R+5$, the algorithm in Section 5, given only (n, C, R, h), succeeds with probability at least $1 - 2^{-h}$ using independent fair bits. It returns some verified representation of the form (1), with

nonzero coefficient and support dimension at most R. It uses at most $N = 8 \cdot 2^{4R}h$ trials and $(4n + s)N$ fair bits. On every tape it terminates with a verified representation or explicit failure. Every accepted representation is sound even if the sufficient promise fails. The conservative bit-time and working-space bounds are

$$\widetilde{O}(2^{4R}hn^{12}), \qquad \widetilde{O}(n^8 + \log(h+1)), \tag{2}$$

where the soft notation hides logarithmic factors in n and $h + 1$.

The theorem does not assert uniqueness of K, f, c, or U_*. In particular a valid alternative representation is a successful output. Malformed input is rejected before any random draw. Failure on a well-formed input is not a certificate that no compatible representation exists.

2 Four contractions expose the scalar field

For this section identify V with K^3 through the unknown f and write $C_0(x, y, z) = \mathrm{Tr}(c \det(x, y, z))$. A useful elementary fact is that $(x, y) \mapsto \mathrm{Tr}(\lambda xy)$ is a nondegenerate binary pairing for every $\lambda \neq 0$. Indeed the trace polynomial $X + X^2 + \cdots + X^{2^{s-1}}$ is nonzero of degree less than 2^s, so the trace functional is nonzero; multiplication by any nonzero element is bijective.

Lemma 2 (Four-contraction identification)**.** *Let v_1, v_2, v_3 be a K-basis and let $v_4 = av_1 + bv_2 + dv_3$. If at least one of a, b, d generates K as an $\mathbb{F}_2$-algebra, then the binary vector space*

$$\mathcal{B} = \{T \in \mathrm{End}_{\mathbb{F}_2}(V) : T^{\mathsf{t}} B_j = B_j T,\ j = 1, 2, 3, 4\}, \qquad B_j(x, y) = C_0(x, y, v_j),$$

is exactly the s-dimensional space of scalar multiplications by K.

Proof. The radical of B_j for $j \leq 3$ is Kv_j: if x is outside that line, the map $y \mapsto \det(x, y, v_j)$ is a nonzero K-linear functional onto K, and the trace pairing is nondegenerate. A balanced endomorphism preserves each radical, since $B_j(Tx, y) = B_j(x, Ty) = 0$ for x in the radical. It is therefore diagonal in this three-block decomposition, with binary maps $A_1, A_2, A_3 : K \to K$.

Use the pairing $\langle x, y\rangle = \mathrm{Tr}(\lambda xy)$, where $\lambda = c \det(v_1, v_2, v_3) \neq 0$. Balancing the three forms on their off-diagonal blocks gives $A_2^* = A_3$, $A_1^* = A_3$, and $A_1^* = A_2$. Consequently $A_1 = A_2 = A_3 = A = A^*$. In the fourth form, the three off-diagonal pairings are weighted by a, b, d. Their balance equations say that A commutes with multiplication by each of these coefficients. A map commuting with a field generator commutes with all its polynomial expressions, so is K-linear. On the one-dimensional K-space K, it is multiplication by a scalar. Conversely, scalar multiplication balances all four forms by K-multilinearity of the determinant. This proves equality. □

Three contractions alone impose self-adjointness of A but not K-linearity. The space of self-adjoint maps has binary dimension $s(s + 1)/2$, because composition with a fixed nondegenerate pairing identifies it with symmetric binary matrices. Thus the fourth equation carries necessary additional information for this argument.

3 Recovering the defect with two translates

For a tensor E let its third mode image be

$$W(E) = \mathrm{span}_{\mathbb{F}_2}\{E(x, y, \cdot) : x, y \in V\} \subseteq V^*.$$

For alternating E, symmetry makes this the mode image in every argument. It is the smallest support space: if $z \in W(E)^{\perp}$, every $E(x, y, z)$ vanishes, and symmetry gives the corresponding statements in the other arguments. The tensor therefore descends to $V/W(E)^{\perp}$ in all arguments. Conversely every support space contains all contracted covectors.

For a candidate scalar action T, define

$$D_T C(x, y, z) = C(Tx, y, z) + C(x, Ty, z). \tag{3}$$

If T is multiplication by $t \in K$, then $D_T C_0 = 0$.

Lemma 3 (Two-translate image recovery)**.** *If E is supported on U and $U \cap T^*U = 0$, with T invertible on U in the sense that $\dim T^*U = \dim U$, then $W(D_T E) = W(E)$. Equivalently it suffices that a basis for U, followed by its T^*-translate, is linearly independent. The empty support case also holds.*

Proof. Write $E(x, y, z) = e(u(x), u(y), u(z))$, where $u : V \to \mathbb{F}_2^r$ records a basis of U. The joint map $x \mapsto (u(x), u(Tx))$ is onto, because its $2r$ coordinate covectors are independent. Every contracted covector of $D_T E$ is a sum of two contracted covectors of E, proving one inclusion. For any x, y, choose a with $u(a) = 0$ and $u(Ta) = u(x)$. Then

$$D_T E(a, y, z) = e(u(x), u(y), u(z)) + e(0, u(Ty), u(z)) = E(x, y, z)$$

for all z. This proves the reverse inclusion. When $r = 0$ both tensors vanish. □

The algorithm needs only $W(D_T C)$, obtained by binary elimination on explicit coefficient slices. It does not test cleanness against a supplied U_*. Under the good event this recovered space is $W(E) \subseteq U_*$. An alternating tensor supported on an r-dimensional space is a linear combination of its $\binom{r}{3}$ exterior basis forms. This follows by choosing dual coordinates on the quotient: alternation sets repeated indices to zero, and the strictly increasing triples specify all remaining coefficients.

4 Conditional sampling and amplification

Put $q = 2^s$, $r = \dim U_*$, and $A = U_*^{\perp}$. Four independent uniform vectors in V all lie in A with probability $2^{-4r} \geq 2^{-4R}$. Conditional on this event they are independent uniform elements of A; their field coordinates need not be uniform in K.

The first three fail to form a K-basis with probability at most

$$\frac{1 + q + q^2}{|A|} = 2^r \frac{1 + q + q^2}{q^3}.$$

At the successive stages the forbidden sets are the origin, a K-line, and a K-plane. Intersecting with A only decreases their sizes. Conditional on any selected basis triple, the first coordinate of v_4 has fibers of size at most q^2 within A.

Define

$$\tau(s) = \sum_{d \mid s,\ d < s} 2^d.$$

A non-generator has minimal polynomial degree a proper divisor of s and belongs to the corresponding proper subfield. Hence there are at most $\tau(s)$ non-generators. Every proper divisor is at most $\lfloor s/2 \rfloor$, so $\tau(s) \leq \sum_{d=1}^{\lfloor s/2 \rfloor} 2^d < 2^{\lfloor s/2 \rfloor + 1}$. The conditional chance that the first coordinate of

v_4 is not a generator is at most $2^r\tau(s)/q$. Therefore the clean quadruple identifies the field with conditional probability at least $1-\beta$, where

$$\beta = 2^R\left(\frac{1+q+q^2}{q^3}+\frac{\tau(s)}{q}\right). \tag{4}$$

This bound holds for every earlier basis triple; it makes no independence assumption about the field coordinate and A.

After computing any ordered binary basis of the correct balance space, an independent uniform s-bit combination is a uniform scalar $t \in K$. Identify V^* with K^3 using the nondegenerate coordinatewise trace pairing. Under this identification T^* multiplies by t. Dependence of $U_* + tU_*$ yields a nonzero pair $(u_0, u_1) \in U_*^2$ with $u_0 + tu_1 = 0$. If $u_1 = 0$ there is no root. Otherwise one nonzero K-coordinate of u_1 determines at most one t. A union bound over the $2^{2r}-1$ nonzero pairs thus bounds bad translate position by $(2^{2r}-1)/q$. Joint generation and good position have probability at least

$$\gamma = 1 - \frac{\tau(s)}{q} - \frac{2^{2R}-1}{q}. \tag{5}$$

If $s \geq 2R+5$, then

$$2^R\frac{1+q+q^2}{q^3} < 2^{R+1-s} \leq \frac{1}{16}, \qquad 2^R\frac{\tau(s)}{q} < 2^{R+1-\lceil s/2\rceil} \leq \frac{1}{4}.$$

Thus $\beta < 1/2$. Also $\tau(s)/q < 1/4$ and $(2^{2R}-1)/q < 1/32$, so in particular $\gamma \geq 1/4$. Each trial succeeds with probability at least $2^{-4R}(1-\beta)\gamma \geq 2^{-4R-3}$. All five draws in each trial are fresh, including the scalar draw in trials that will fail field identification. With $N = 8\cdot 2^{4R}h$, independence gives

$$\Pr(\text{all trials fail}) \leq (1-2^{-4R-3})^N \leq e^{-h} \leq 2^{-h}.$$

Earlier acceptance of a different verified representation only helps.

5 Algorithm and exact certificate

For each of the prescribed N trials perform the following operations.

1. Draw four n-bit vectors and an s-bit scalar index. Construct their contractions directly from the supplied coefficients.

2. Solve the simultaneous balance equations on n^2 unknown bits. Reject unless the solution space has dimension s and contains the identity.

3. Use the scalar index to select T in this space. Form $I, T, \ldots, T^s$. Require $I, \ldots, T^{s-1}$ independent and in the balance space, and solve $T^s = \sum_{j<s} p_j T^j$. Test irreducibility of $p(X) = X^s + \sum_{j<s} p_j X^j$. Reject on failure.

4. In $K = \mathbb{F}_2[X]/(p)$, greedily choose three module generators from the physical binary basis. For each chosen vector append its first s iterates under T. Require the resulting n columns to be independent and invert this matrix to obtain $f : V \to K^3$.

5. Compute $U = W(D_T C)$, and reject if its dimension exceeds R. Construct the s trace-determinant coefficient columns in these coordinates, and the $\binom{\dim U}{3}$ exterior-cube columns on a basis of U. Solve for the coefficients representing the whole tensor C.

6. Reject if there is no solution or the compatible coefficient is zero. Independently verify the complete serialized certificate, and accept only if that check passes. After N rejected trials, return failure.

The irreducibility test can compute successive squares of X modulo p, requiring $X^{2^s} = X$ and $\gcd(p, X^{2^k} - X) = 1$ for every $1 \le k \le \lfloor s/2 \rfloor$. For completeness of this test, a reducible degree-s polynomial has an irreducible factor of degree at most $s/2$; such a factor divides $X^{2^k} - X$ for its degree k. An irreducible degree-s polynomial satisfies all the required tests. No factoring oracle or table of field elements is used.

The relation and independence tests make $\mathbb{F}_2[T]$ a field of degree s. Each nonzero vector in the quotient by previously selected K-lines generates another s-dimensional binary summand; otherwise a nonzero field scalar would annihilate its nonzero quotient class. Greedy choice therefore constructs a K-basis in three steps when the field tests pass. Direct verification of the resulting binary basis is retained.

The certificate contains p, the $n \times n$ binary coordinate map f, a nonzero s-bit c, independent support covectors $u_1, \dots, u_r$ with $r \le R$, and $\binom{r}{3}$ binary exterior coefficients. Its size is $O(n^2 + nR + R^3 + s)$ bits. The verifier rechecks irreducibility, ranges, independence, and all $\binom{n}{3}$ coefficients of (1). Both summands are alternating trilinear by construction, so checking these coefficients proves equality on every triple of physical vectors. This is exact certificate soundness; the verifier never trusts a claimed clean event.

Under a good trial, Lemmas 2 and 3 supply the correct field and the minimal error support. The full coefficient system is therefore consistent. Moreover its compatible coefficient cannot accidentally be zero: a nonzero trace-determinant tensor has mode image all of V^*, by the trace nondegeneracy argument, whereas a tensor supported on U has mode image contained in U. Since $\dim U \le R < n$ under the stated field-size condition, the compatible tensor space intersects the supported error space only in zero. Thus any solution to the coefficient system has the same nonzero compatible component in the recovered coordinates. This completes the correctness and probability proof of Theorem 1.

6 Complete resource accounting

All field operations use s-bit polynomial representatives. Schoolbook multiplication and reduction use $O(s^2)$ bit operations; trace evaluation uses s squarings and $O(s^3)$ bit operations. Binary vectors and matrices are stored explicitly. The following deliberately loose bounds use elementary elimination, without assuming unit-cost operations on unbounded integers.

Four contractions require $O(n^3)$ bit operations. The balance solve has $O(n^2)$ rows and n^2 variables, so uses $O(n^6)$ bit operations and $O(n^4)$ bits of storage. Forming matrix powers, checking their spans, and constructing and inverting a module basis each fit within $O(n^6)$ bit operations and $O(n^4)$ storage. The successive-squaring and gcd irreducibility procedure is polynomial and fits within the same envelope. Computing the defect slices and their image costs $O(n^4)$ with dense binary matrix arithmetic and elimination.

The coefficient fit has $d = O(n^3)$ rows and $m = s + \binom{r}{3} = O(n^3)$ columns, even on failed trials with $r \le R \le n$. Constructing the compatible columns costs $O(n^7)$ bit operations using the trace routine just described. Constructing the exterior columns costs $O(n^6)$; elimination costs $O(dm \min(d, m)) = O(n^9)$ and stores $O(dm) = O(n^6)$ bits. Certificate rechecking reconstructs these columns and costs no more. All stages can discard one failed candidate before starting the next. Polynomial index and interpreter representation overheads are absorbed conservatively by

$\widetilde{O}(n^{12})$ time and $\widetilde{O}(n^8)$ working space per trial. These are envelopes, not optimized or empirically measured scaling laws.

The trial counter uses $O(R + \log(h+1))$ bits; its arithmetic, input validation, reading and storage of C, and serialization of the certificate are included. Incremental counter work is at most $O(N(R + \log(h+1)))$. Storage has no factor N. The implementation retains a fixed number of rejection counters, not all previous candidates. Each trial consumes exactly $4n + s$ bits, even if an early algebraic test fails. Multiplying the envelope by N gives (2). At $R = O(\log n)$ and polynomial h, this is polynomial in the dense input size. The exponential dependence on R remains explicit.

7 Comparison with established methods

Hora's ring forms [1] include composing a determinant over a characteristic-p ring with a linear functional. His results study radicals, decomposition and equivalence, including characteristic-two examples. Thus the uncorrupted form used here is established mathematical structure. The recovery theorem adds an unknown common support and a bounded randomized constructor with exact verification; it does not originate ring forms.

Computational tensor invariants already use adjoints, nuclei and centroids to expose underlying scalar structure. The Magma documentation [2] explicitly includes underlying-field recovery and points to Brooksbank–Wilson [3] for cyclic-algebra machinery. Their module isomorphism theorem concerns isomorphisms over rings and includes an efficient cyclic-ring case. Our simultaneous balance solve is elementary linear algebra within that established approach. The theorem here specifies when four corrupted contractions happen to identify the field, then certifies the full input identity. We do not assert a general tensor-isomorphism algorithm.

Fouque–Granboulan–Stern [4] exploit differential kernel dimensions in multivariate cryptanalysis, including internally perturbed Matsumoto–Imai systems. This is a direct precedent for finding directions that cancel a hidden perturbation. Their cryptographic input and inversion task differ from a scalar-valued alternating cubic tensor and its compatible representation certificate.

Dubois–Granboulan–Stern [5], Theorem 1, analyzes a random instance of internally perturbed HFE and relates derivative-kernel distributions to random linear maps for a large proportion of instances. Their recovery uses a statistical distinguisher for the perturbation kernel. In contrast, our randomness belongs to the algorithm and the corruption tensor is arbitrary under its support promise. This distinction in quantifiers is essential; their theorem is not a stated worst-case guarantee for our tensor class. These comparisons delimit the proposed contribution. They are a focused primary-source screen, not an exhaustive novelty or priority proof.

8 Reproducibility, formal status, and limitations

The companion at Quantyra/corrupted-field-recovery contains `src/recovery.py` and `tests/check_recovery.py`. Run the latter with Python 3.13; it writes an environment and source-hash receipt to `artifacts/recovery-checks.json`. The tests distinguish ordinary seeded tapes from fixture-assisted tapes. They recreate 200 exact rational probability-bound checks, nine four-contraction field identifications, and 104 two-translate image checks. These counts are saved executable evidence, not the unsaved earlier conversation checks.

The suite also exercises coordinate changes, independent pointwise certificate evaluation, altered certificates and tensors, malformed inputs, a smaller rank-cap refusal, and all 16 trials of an $R = 0, h = 2$ zero tape. The $s = 11, R = 3$ scripted case satisfies the sufficient field-size promise but is not an ordinary-random success-frequency measurement. The $s = 3, R = 3$ ordinary seeded

positive has covectors in three independent K-directions and is outside that sufficient condition. A first attempted $s = 3, R = 3$ fixture instead filled one entire K-coordinate; its seeded run exhausted the bound and is retained as a development failure, not silently reported as a positive test. The $s = 5$ negative control checks that two three-dimensional subspaces of one five-dimensional K-line cannot be independent, for all 32 scalars.

Lean sources and a theorem-to-obligation inventory accompany this draft. Compiling individual lemmas is not complete certification of this manuscript. The four-contraction field theorem, finite-field cardinality and sampling bridge, complete bounded constructor, verifier equivalence and bit-cost formalization must all be discharged before a complete certification claim. The Python receipt alone establishes none of these formal obligations.

The origin is R05-08 at historical revision `d2ae8c6fc275423c2276ba82644b8993660ec684` in `Quantyra/quantyra-quantum-complexity`; exact paths are recorded in `PROVENANCE.md`. The present theorem isolates cubic tensors. The broader phase-recovery application needs additional quadratic and pairing steps and retains its separate $s \geq 7R+5$ sufficient condition. No claim about complete circuit simulation, quantum advantage, counting backends, or BPP versus BQP follows by replacing that condition with $2R + 5$.

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